

Brandon Koehn

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Education

B.S. in Environmental Biology (GPA: 3.76) Expected Spring 2026

California State Polytechnic University Pomona, CA

CSU-SPaRA Research Scholar

A.D.T Environmental Science | A.D.T Biology | A.S. Forestry | A.S. Biology 2022 –2024

Citrus College, CA

Skills & Qualifications

Diving Certification: Padi Open Water Diver Certified Dec. 2025

- 6 logged dives | Max depth: 10.6 m | Total bottom time: ~2 hrs 10 min

Technical Skills: Marine field survey methods (Snorkeling), R programming (Intermediate), Policy Brief Writing, Research Proposal Development

Laboratory Skills: Epifluorescence microscopy, Colorimetric assays, Microbial/Diatom Culturing, Invertebrate Dissections

Research Experience

Undergraduate Researcher, Bonisoli Alquati Lab 2025 – Present

California State Polytechnic University, Pomona

Undergraduate Project: “**Assessing Invertebrate Condition & Kelp Distribution Between the Santa Clara and Los Angeles River**”

- Conducting research contributing to a collaborative project investigating contaminant exposure across agricultural and urban watersheds
- Collecting and dissecting marine invertebrate and kelp specimens to support chemical analysis of pesticide load
- Evaluating organism health condition indices for *Mytilus californianus*, and *Macrocystis pyrifera*
- Mapping spatial distribution of *Macrocystis pyrifera* relative to river outflows using the `simple features` package in R
- Performing statistical analysis and data visualizations in R

Dr. Jayson Smith Costal Ecology Lab, Biological Sciences Department 2025 – Present

California State Polytechnic University, Pomona

Volunteer Research Assistant

- Assisted graduate students and MARINe scientists with rocky intertidal ecological monitoring and field projects

- Collected data using standardized sampling methods including line transects and quadrat surveys
- Identified native algal and invertebrate species to support biodiversity documentation
- Completed over 60 hours of ecological field research experience

BIO 4550/L Tropical Marine Ecology, Panama

2026

Project: “No Jelly Left Behind: *Cassiopea* Habitat Selection & Host Preference of a Parasitic Nudibranch”

- Surveyed *Cassiopea* spp. at 3 different locations to determine habitat selection and host preference of the parasitic nudibranch *Dondice parguerensis*
- Aided in a sea slug biodiversity survey across Bocas del Toro for 5 days, stationed at ITEC.

USC Wrigley Institute REU in Coastal Ocean Processes

2025

USC-MEB Dr. David Hutchins Lab | *Mentor:* Bradley Mackett

Research Project: “Wildfire-Derived Nutrients as Possible Drivers of *Pseudo-nitzschia* Blooms in The San Pedro Channel”

- Investigated the effects of wildfire runoff (ash and Phos-Chek) on the physiology and toxicity of *Pseudo-nitzschia*
- Examined how wildfire runoff may impact natural phytoplankton (*Chaetoceros* spp.) and zooplankton communities through an overnight feeding experiment with Copepod and Cladocera taxon.

Pathways to STEM: Experiential Learning Program

2023 – 2024

Vermont Module | Environmental Sciences | 1-week

Research Project: “Wetland Rehabilitation for Sustained Life”

- Developed a policy brief on freshwater conservation through the lens of Civic Science Education
- Achieved through literature review on the benefits of beaver restoration and removal of man-made dams

Catalina Module | Marine Biology | 3-days

Research Project: “Microbial Abundance at Varying Depths”

- Collected seawater samples at multiple depths and conducted epifluorescence microscopy counts to differentiate phytoplankton, bacteria, and viruses
- Developed and applied core communication strategies by tailoring rhetorical approaches for both scientific peers and public audiences

Oak Crest Institute of Science Module | Clinical Research | 6-weeks

Research Project: “Mean Volume of Cheek Cells Using the Nucleator Method”

- Swabbed the cheeks of 10 volunteers, embedding cells in agarose, and captured confocal images to determine the mean cheek-cell volume using the nucleator stereological method
- Gained hands-on experience with biosafety level 2 (BSL-2) protocols and safe handling of human-derived samples

CU Boulder Module | Climate Science | 1-week

Research Project: **“Bioaccumulation of Mercury in Freshwater Game Fish”**

- Used publicly available NOAA and EPA datasets to quantify methylmercury accumulation in Rainbow Trout and Largemouth Bass over time.
- Used R and RStudio for geospatial analysis and trend visualization

Relevant Experience

Peer Mentor | Pathways to STEM 2025

- Lead 3-day research project sampling seawater at varying depths to analyze the concentration of orthophosphate using a colorimetric
- Mentored 4 *community college* students during project, teaching them to conduct the orthophosphate assay, and interpret their data as part of a collaborative marine science project.

Student volunteer | Monrovia Canyon Park 2023 – 2024

- Aided in trail maintenance, invasive species removal, and workshop maintenance for over 120 hours

Honors & Awards

COAST Undergraduate Research Funding Award (\$750)	2026
CSU-SPaRA Research Scholar (NSF SPIRES-funded) (\$2,200)	2025 – 2026
President’s Honors List, Cal Poly Pomona	2025 – 2026
Dean’s Honors List, Cal Poly Pomona	2024 – 2026
NSF REU Stipend USC Wrigley Institute (\$6,000)	2025
Wildland Resources & Forestry Scholarship (\$500)	2023
Certificate of Recognition Wildland Resources and Forestry	2023
Certificate of Recognition Economics	2023

Presentations & Outreach

Poster Presentation, “Wildfire-Derived Nutrients as Possible Drivers of *Pseudo-nitzschia* Blooms in The San Pedro Channel”, SoCal SETAC Annual Meeting, Apr. 2026

Poster Presentation, “Assessing Mussel Condition and Giant Kelp Distribution Between the Santa Clara and Los Angeles River”, College of Science Symposium, Cal Poly Pomona, Apr. 2026

Poster Presentation, “No Jelly Left Behind: *Cassiopea* Habitat Selection & Host Preference of a Parasitic Nudibranch””, College of Science Symposium, Cal Poly Pomona, Apr. 2026

Web Article, USC Wrigley Institute for Environment and Sustainability Blog showcased “[Ash to Algae: Tracing the Connection between Wildfires and Harmful Algal Blooms](#)”, Nov. 2025

Poster Presentation, “Wildfire-Derived Nutrients as Possible Drivers of *Pseudo-nitzschia* Blooms in The San Pedro Channel”, Annual SoCal Microbiome Symposium, California Institute of Technology, Sep. 2025

Student Panelist, USC Young Researchers Program (USC-YRP), Jul. 2025

Oral Presentation, “Wetland Rehabilitation for a Sustained life” Public talk in Marlboro, Vermont, Aug. 2024

Poster Presentation, “Bioaccumulation of Mercury in Freshwater Game Fish”, University of Colorado Boulder, Jun. 2023

Affiliations

Member , Society of Environmental Toxicology & Chemistry	2026 – Present
Member , SACNAS	2025 – Present
Scholar , California State University STEM Pathways and Research Alliance	2025 – Present